

AP CSP | Lesson 21: Robot Algorithms

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Complete all 12 original questions (30 marks, about 40 minutes). Q5 requires exactly TWO choices. Trace statements in their execution order. Each program statement and each brace occupies its own line. Show reasoning and use the stated valid input range.

1. A robot faces east. What is its heading after ROTATE_LEFT()? [1 mark]

A. North B. South C. East D. West

Answer: _____

2. A robot at (row 4, column 1) faces east. The next cell is open. After MOVE_FORWARD(), where is it? [1 mark]

A. (3,1) B. (4,2) C. (5,1) D. (4,1)

Answer: _____

3. A robot evaluates CAN_MOVE(left). Which statement is correct? [1 mark]

A. It moves left if that cell is open. B. It rotates left before checking.
C. It returns a Boolean without moving or rotating. D. It returns the absolute direction west.

Answer: _____

4. According to the AP robot model, what happens if MOVE_FORWARD() would enter a blocked cell? [1 mark]

A. The robot jumps over the wall. B. The robot rotates right.
C. The move is skipped and all later commands execute. D. The robot stays in its cell and the program terminates.

Answer: _____

5. Which TWO statements are correct? Select exactly TWO. [1 mark]

A. Rotations take place without changing cells. B. CAN_MOVE(forward) uses the top of the diagram regardless of heading.
C. Three right turns and one left turn produce the same final heading. D. Entering a marked goal cell always terminates the AP program.

Answer: _____

6. All neighboring cells are open. Starting at (4,1,E), execute ROTATE_LEFT() then MOVE_FORWARD(). What is the final state? [1 mark]

A. (4,2,N) B. (4,1,N) C. (3,1,N) D. (5,1,S)

Answer: _____

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Written responses | Questions 7-9

7. [4 marks]

A robot starts at (row 4, column 1), facing east. All visited cells are open. Execute `MOVE_FORWARD()`, then `ROTATE_LEFT()`, then `MOVE_FORWARD()`. State the position after the first command, the position and heading after the second, and the final state.

8. [4 marks]

A robot at (4,2) faces north. Cell (3,2) is blocked; (4,1) and (4,3) are open. State `CAN_MOVE(forward)`, `CAN_MOVE(left)`, and `CAN_MOVE(right)`. State whether evaluating these calls changes position or heading.

9. [4 marks]

Use a 5×5 grid, rows top-to-bottom 1-5 and columns left-to-right 1-5. Start (4,1,E); walls are (3,2),(3,3),(2,2). Write a counted-loop program that moves east 3 cells, turns left, then moves north 3 cells. State its final state. Each brace occupies its own line.

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Written responses | Questions 10-12

10. [4 marks]

A robot faces a blocked cell. Compare these programs: A executes `MOVE_FORWARD()` then `ROTATE_RIGHT()`. B first tests `CAN_MOVE(forward)`, executes `MOVE_FORWARD()` only when that test is true, then executes `ROTATE_RIGHT()` after the IF. For each program, state whether a move occurs and whether the final rotation executes.

11. [4 marks]

A correct route uses six forward moves and one left turn. Another uses the same movements but replaces the left turn with three right turns at the same point. State whether final position and heading agree, count executed turns for both, and explain which has fewer executed movement-or-turn commands.

12. [4 marks]

A robot reaches (1,4,N) on a 5×5 grid, the marked goal. Its program then has one more `MOVE_FORWARD()` followed by `ROTATE_RIGHT()`. Explain whether the goal marker stops it, whether the next move is legal, where it remains, and whether the rotation executes.
